

Systematic review handover

Intervention review · 16 included studies · 4,476 records screened · PRISMA 2020 · Cochrane RoB 2 · GRADE

This is a redacted sample. It shows the structure and level of detail of a real deliverable. Client names, journal names and identifying particulars have been removed or changed. Ask us and we will send one built around a review like yours.

1 Protocol summary

Registered on PROSPERO before screening began. Two deviations from the registered protocol arose during the review and are declared in the manuscript: one subgroup was dropped because too few trials reported it, and the follow-up window for the primary outcome was fixed at twelve months after two trials were found to report at both six and twelve.

Element	As registered
Framework	PICO
Population	Adults with chronic heart failure, any ejection fraction, diagnosis confirmed
Intervention	Supervised or home-based exercise training of at least eight weeks
Comparator	Usual care, or an intervention with no exercise component
Primary outcome	All-cause hospital readmission within twelve months
Secondary outcomes	All-cause mortality, six-minute walk distance, health-related quality of life, serious adverse events
Eligible designs	Randomised controlled trials, including cluster and cross-over designs
Subgroups	Supervised against home-based delivery; preserved against reduced ejection fraction
Language and date	No language restriction; no date restriction
Reporting guideline	PRISMA 2020, with PRISMA-S for the search
Appraisal instrument	Cochrane RoB 2
Certainty framework	GRADE, applied per outcome

2 Search strategy, as run

Reproduced here for MEDLINE (Ovid). The strategy was translated line by line for each other platform rather than pasted across, because Emtree and MeSH do not share headings and the proximity syntax differs. Every version is supplied with the handover, dated, with the record count each line returned.

#	Line	Records
1	exp Heart Failure/	142,318

#	Line	Records
2	(heart adj2 failure or cardiac failure or HF _r EF or HF _p EF).ti,ab,kf.	189,047
3	1 or 2	214,806
4	exp Exercise Therapy/ or exp Cardiac Rehabilitation/	88,412
5	(exercis* or train* or rehabilitat* or physical activity).ti,ab,kf.	1,204,558
6	4 or 5	1,229,730
7	3 and 6	6,904
8	7 and Cochrane highly sensitive RCT filter (sensitivity-maximising version)	1,842

Databases searched: MEDLINE (Ovid) 1,842 · Embase 2,310 · Cochrane CENTRAL 1,104 · CINAHL 486 · Web of Science 1,376 · ClinicalTrials.gov and WHO ICTRP 212 · citation searching 64. All searches run on the same date and re-run before submission.

3 PRISMA 2020 flow

Stage	n
Records identified from databases	7,118
Records identified from registers and grey literature	212
Records identified by citation searching	64
Duplicates removed before screening	2,918
Records screened on title and abstract	4,476
Records excluded at screening	4,389
Reports sought for retrieval	87
Reports not retrieved	5
Reports assessed for eligibility at full text	82
Reports excluded with reasons	66
Studies included in the review	16
Reports of included studies	22

Full-text exclusions, with reasons

Reason	n
Wrong population — mixed cardiac cohorts with no heart-failure subgroup	21
Wrong intervention — education or diet only, no exercise component	16
No readmission outcome reported	13

Reason	n
Conference abstract with no full report	9
Secondary report of a trial already included	7

The full-text exclusion list is a required part of the published review. It is compiled as screening proceeds, because it cannot be reconstructed afterwards.

4 Data extraction and risk of bias

Every study was extracted independently by two reviewers into a form piloted on four studies, then reconciled. Eleven disagreements arose; all were resolved by discussion, and three required the corresponding author to be contacted. Where a number was obtained from an author rather than from the published report, its provenance is recorded in the extraction table.

Study	Design	RoB 2	Note
Multicentre RCT · United States, 2009	RCT	Some concerns	Largest trial; drives the pooled estimate
Single-centre RCT · Germany, 2004	RCT	High	Deviations from intended intervention
Multicentre RCT · Italy, 2013	RCT	Low	—
Nurse-led RCT · Australia, 2010	RCT	Some concerns	Indexed in CINAHL only alongside MEDLINE
Home-based RCT · Netherlands, 2019	RCT	Low	—
Pragmatic RCT · Belgium, 2021	RCT	Some concerns	Indexed in Embase only
Multicentre RCT · China, 2018	RCT	Some concerns	Published in Chinese; translated for extraction
Registry-posted RCT · Canada, 2023	RCT	High	Results posted to registry, never published

Extract only — eight of sixteen studies shown. Overall: four at low risk, nine with some concerns, three at high risk. The high-risk trials are removed in a pre-specified sensitivity analysis, and the pooled estimate holds.

Two findings worth noting from the search. One included trial is indexed in Embase and in no other database searched, and one exists only as results posted to a trial registry. A review of this question built on MEDLINE alone would have missed both, and nothing in its finished manuscript would have shown that it had.

5 Summary of findings (GRADE)

Outcome	Studies	Certainty	Reason for downgrade
All-cause readmission, 12 months	11 (2,847)	Moderate	Risk of bias — three trials at high risk

Outcome	Studies	Certainty	Reason for downgrade
All-cause mortality, 12 months	9 (2,208)	Low	Risk of bias; imprecision — interval spans benefit and harm
Six-minute walk distance	12 (3,014)	Moderate	Risk of bias
Health-related quality of life	8 (1,942)	Low	Inconsistency; indirectness — four instruments, two unvalidated in heart failure
Serious adverse events	5 (1,106)	Very low	Not pooled — no shared definition; reported narratively under SWiM

Heterogeneity for the primary outcome was low and the direction of effect was consistent across every trial that measured it. Publication bias was assessed by funnel plot and Egger test for the primary outcome only, since fewer than ten trials reported each of the others.

6 What is handed over

- The manuscript in IMRaD, with a PRISMA 2020-compliant structured abstract, formatted to the target journal's instructions for authors.
- The completed PRISMA 2020 checklist and the flow diagram as an editable figure.
- The search strategy for every database, line by line, with record counts and the dates run, reported to PRISMA-S.
- The screening and extraction spreadsheets, including the full-text exclusion list with reasons.
- The risk-of-bias assessments per study and per domain, as a traffic-light figure and as data.
- The analysis files — R, Stata or RevMan — so the synthesis can be re-run rather than trusted.
- The PROSPERO record and a written note of every deviation from it.

Pubrica supports the work; authorship remains yours. Where our contribution should be acknowledged under ICMJE, the handover drafts the wording for you to approve.

Prepared by Pubrica. Pubrica is a trading name of Guires, registered in England, Wales and India. pubrica.com